

BAYANDAY, Russian FederationWMO: **306270**Lat: **53.100N**Lon: **105.533E**Elev: **2498**StdP: **13.42**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.8	-28.5	-37.4	0.7	-32.4	-32.9	0.9	-28.5	13.4	-4.3	11.9	-7.9	4.4	30	0.629

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	82.3	63.5	78.8	62.0	75.6	60.8	66.4	76.8	64.5	73.6	62.8	72.4	5.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	94.3	70.2	61.1	88.3	68.9	59.4	82.8	67.3	32.6	77.3	31.1	73.7	29.8	72.7	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	12.8	10.8	9.5	DB	-35.9	87.9	8.1	4.0	-41.8	90.8	-46.6	93.1	-51.2	95.4	-57.1	98.3
(5)				WB	-35.9	69.5	7.8	2.8	-41.5	71.5	-46.1	73.1	-50.5	74.7	-56.2	76.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	30.4	-6.7	1.1	17.6	35.0	47.0	59.5	64.1	59.2	46.5	31.1	10.5	-2.4
	DBStd	26.55	11.98	12.15	11.25	9.84	8.49	7.46	5.26	6.47	8.15	9.67	13.26	12.84
	HDD50	8331	1757	1371	1005	458	160	11	0	8	164	586	1186	1625
	HDD65	12746	2222	1791	1470	901	559	194	79	197	557	1050	1636	2090
	CDD50	1164	0	0	0	7	68	297	437	295	58	2	0	0
	CDD65	103	0	0	0	0	2	30	51	19	1	0	0	0
	CDH74	1052	0	0	0	1	46	353	447	184	21	0	0	0
	CDH80	210	0	0	0	0	7	77	89	35	2	0	0	0
Wind	WSAvg	4.4	4.3	4.1	4.3	5.0	5.7	4.5	4.0	3.9	4.3	4.2	4.2	4.5
Precipitation	PrecAvg	14.8	0.2	0.2	0.2	0.5	1.0	2.3	3.7	3.4	1.7	0.7	0.4	0.4
	PrecMax	20.4	0.6	0.4	0.7	1.8	2.4	5.4	8.8	6.7	4.0	3.4	0.9	1.0
	PrecMin	8.7	0.0	0.0	0.0	0.0	0.0	0.5	1.1	0.7	0.4	0.2	0.1	0.0
	PrecStd	2.5	0.1	0.1	0.2	0.4	0.6	1.1	1.9	1.5	0.9	0.6	0.2	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.4	30.9	47.9	69.0	79.5	86.9	86.5	84.0	77.1	62.4	42.2	27.0
		MCWB	20.6	25.9	37.2	48.6	56.2	65.0	66.3	63.8	59.2	47.3	35.7	23.6
	2%	DB	16.8	26.4	41.7	61.5	72.8	81.7	82.4	78.6	69.7	54.6	35.9	21.3
		MCWB	14.6	22.2	33.2	45.3	53.5	62.7	64.7	62.0	56.0	43.2	30.9	19.3
	5%	DB	12.6	22.3	37.1	55.7	68.3	78.1	78.9	74.8	64.4	49.1	31.7	17.1
		MCWB	11.2	19.0	30.4	42.1	50.6	61.0	63.6	61.2	53.1	40.1	28.3	15.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	9.0	17.8	33.2	50.4	63.3	74.3	75.5	71.2	59.4	44.5	28.5	14.1
		MCWB	7.9	15.3	27.9	39.1	47.7	59.0	62.8	59.6	50.2	37.3	25.9	13.3
	0.4%	WB	21.0	26.0	37.6	49.5	58.2	68.6	70.0	66.1	60.3	48.2	35.8	23.7
		MCDB	22.4	30.3	47.0	66.3	76.3	81.5	80.9	77.0	73.4	61.3	42.0	26.3
	2%	WB	15.0	22.4	33.6	45.7	54.6	64.6	67.5	63.9	57.4	43.8	31.3	19.4
		MCDB	16.9	26.0	41.1	59.6	71.2	76.4	77.9	74.8	68.0	53.4	34.9	21.0
Mean Daily Temperature Range	5% DB	WB	11.4	18.9	30.7	42.6	51.2	62.5	65.6	62.3	54.1	40.6	28.8	15.9
		MCDB	12.6	22.0	36.5	55.3	66.0	74.8	74.9	72.5	62.1	48.7	31.8	17.1
		WB	8.0	15.3	28.1	39.5	48.5	60.3	64.0	60.7	51.4	37.6	26.0	13.2
	5% WB	MCDB	9.0	17.7	33.3	49.5	61.4	71.8	71.8	69.4	58.7	43.8	28.5	14.2
Clear-Sky Solar Irradiance	taub	0.259	0.284	0.312	0.356	0.405	0.431	0.418	0.371	0.325	0.302	0.280	0.259	
	taud	2.105	2.097	2.125	2.254	2.158	2.172	2.256	2.373	2.460	2.428	2.231	2.085	
	Ebn at Noon	224	256	274	277	267	261	262	270	272	253	216	196	
	Edh at Noon	22	31	38	38	44	44	40	34	27	23	20	19	
All-Sky Solar Radiation	RadAvg	385	741	1241	1569	1749	1860	1710	1477	1073	673	378	268	
	RadStd	19	24	49	101	113	104	140	64	96	45	23	20	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	-3.55	N/A	N/A	N/A	N/A	N/A	N/A	N/A